Global X Camera

Functional circuit drawings | Presentation revision of the recovered circuit

Local support circuits use direct wires. Named buses group data and control interfaces.

Global labels join power rails and signals between functional pages. Reference designators and physical pin numbers are unchanged.

This is a layout-only revision. Existing design issues, source assumptions and DNP choices remain for the ongoing electrical review.

2 Battery latch, shutter and status LED

File: 02_battery_latch.kicad_sch

3 Main regulated rails

File: 03_main_power.kicad_sch

4 MCU I/O and analog supplies

File: 04_mcu_io_power.kicad_sch

5 MCU internal SMPS and core supply

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6 Clock, reset, boot and debug

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7 Camera connector, module and CSI-2

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8 Camera power rails

File: 08_camera_power.kicad_sch

9 Boot NOR flash

File: 09_boot_flash.kicad_sch

10 Raw NAND storage

File: 10_nand.kicad_sch

11 USB-C data, protection and supply

File: 11_usb.kicad_sch

12 RTC and accelerometer interfaces

File: 12_rtc_accelerometer.kicad_sch

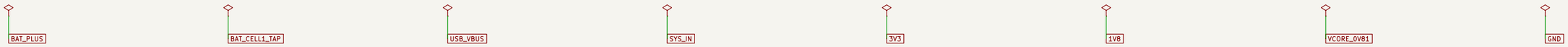
13 MCU grounds and unused GPIO A-D

File: 13_ground_unused.kicad_sch

14 Unused MCU GPIO and interfaces

File: 14_unused_gpio.kicad_sch

Supply-source declarations retained from the recovered schematic for ERC.



Global X Camera

Sheet: /

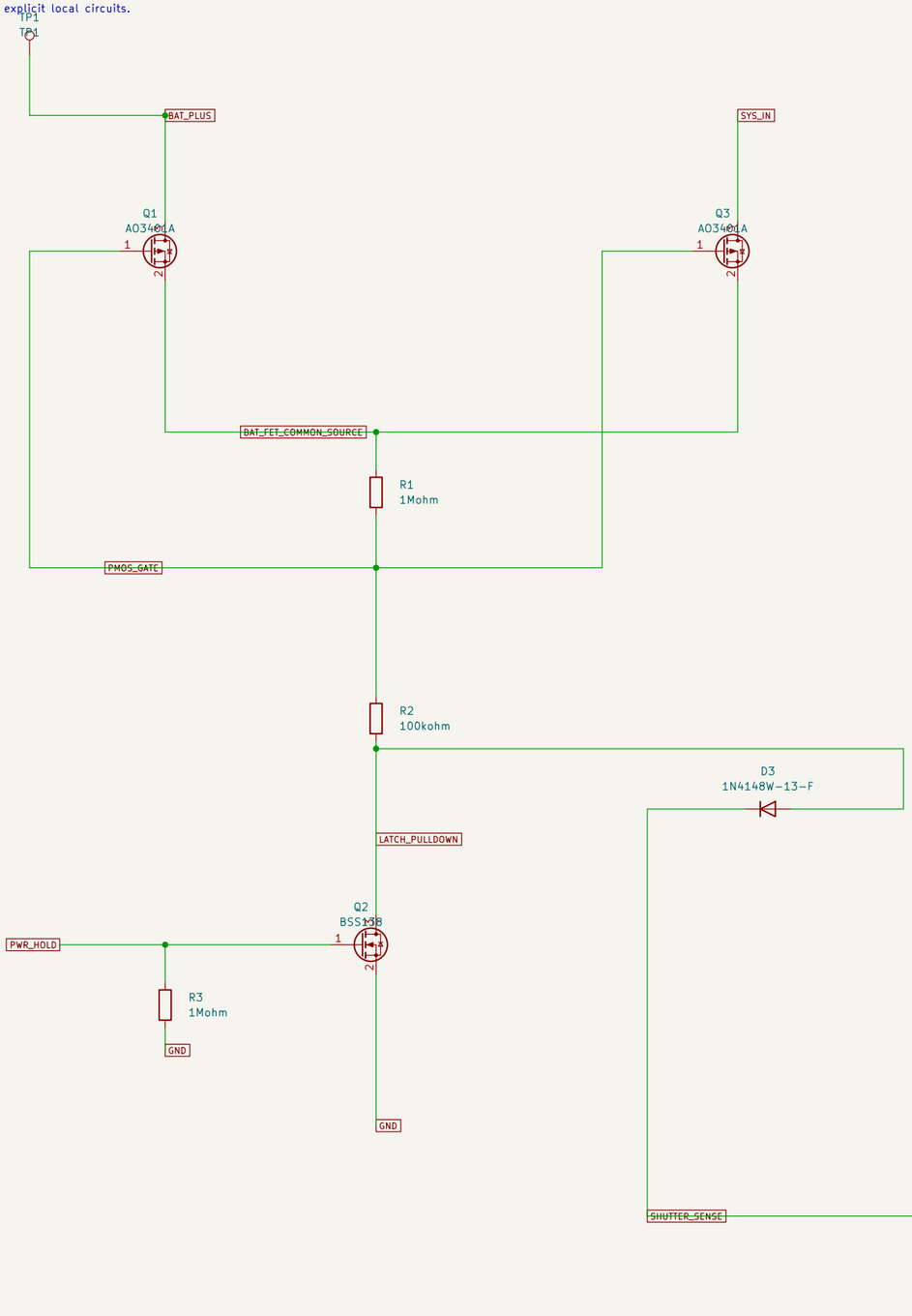
File: global-x-camera-readable.kicad_sch

Title: Global X Camera — readable schematic

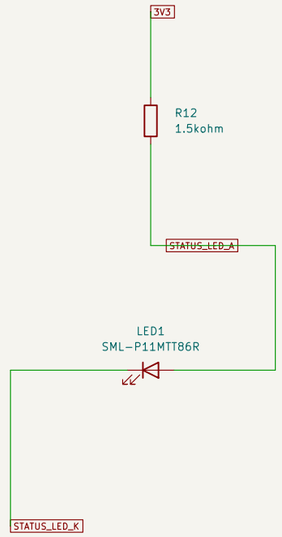
Size: A3	Date: 2026-09-05	**Rev: A-readable**
KiCad E.D.A. 10.0.6		Id: 1/14

Battery latch, shutter and status LED

Original latch topology retained. Gate, source, hold and shutter nets are drawn as explicit local circuits.



Status indicator

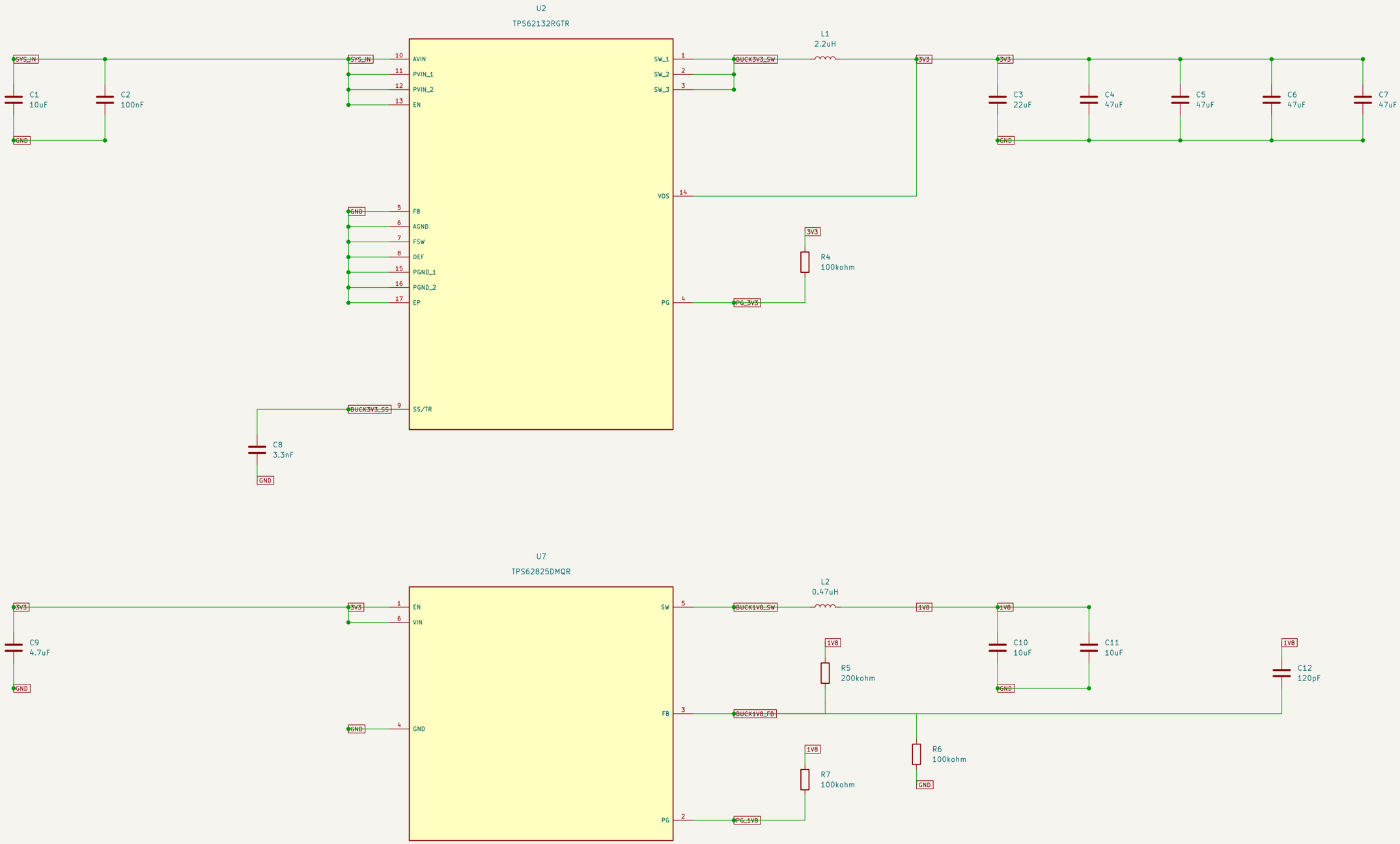


Presentation revision; electrical design review remains pending

Global X Camera		
Sheet: /2 Battery latch, shutter and status LED/		
File: 02_battery_latch.kicad_sch		
Title: Battery latch, shutter and status LED		
Size: A3	Date: 2026-09-05	**Rev: A-readable**
KiCad E.D.A. 10.0.6		Id: 2/14

Main regulated rails

Power flows left to right. Each converter is drawn with its input, output and control network.



Presentation revision; electrical design review remains pending

Global X Camera

Sheet: /3 Main regulated rails/
File: 03_main_power.kicad_sch

Title: Main regulated rails

Size: A3	Date: 2026-09-05	**Rev: A-readable**
KiCad E.D.A. 10.0.6		Id: 3/14

MCU I/O and analog supplies

Supply balls are grouped by rail; bypass capacitors connect directly to the adjacent rail trunk.

U1A

STM32N657X0H3Q

VBAT E1
VDDIO2_1 L13
VDDIO2_2 L14
VDDIO2_3 M14
VDDIO2_4 M15
VDDIO3_1 N13
VDDIO3_2 P13
VDD_1 G13
VDD_2 H12
VDD_3 H13
VDD_4 J12
VDD_5 J13
VDD_6 K13

3V3

C20 100nF
C21 100nF
C22 100nF
C23 100nF
C24 100nF
C25 100nF
C44 100nF
C45 100nF

GND

C46 100nF
C47 100nF
C48 100nF
C49 100nF

VDDA18ADC N6
VDDA18AON F6
VDDA18CSI R5
VDDA18PLL M5
VDDA18PMU G1
VDDA18USB C5
VDDIO4_1 E8
VDDIO4_2 F8
VDDIO5_1 E9
VDDIO5_2 F9
VREF+ U2

1V8

C26 100nF
C27 100nF
C28 100nF
C29 100nF
C30 100nF
C31 100nF
C32 100nF
C33 100nF

C34 100nF
C35 100nF
C60 1uF
C61 100nF

VDD33USB A4 — 3V3_USB

VDDCSI R4 — VCORE_0V81

MCU supplies (every supply ball retained)

USB supply decoupling: USB page. CSI supply uses the existing core rail connection.

Presentation revision; electrical design review remains pending

Global X Camera

Sheet: /4 MCU I/O and analog supplies/
File: 04_mcu_io_power.kicad_sch

Title: MCU I/O and analog supplies

Size: A3	Date: 2026-09-05	**Rev: A-readable**
KiCad E.D.A. 10.0.6		Id: 4/14

MCU internal SMPS and core supply

Core rail bypassing and switch-node components are shown beside the MCU power unit.

U1B

STM32N657X0H3Q

VDDCORE_1 L6
VDDCORE_2 L12
VDDCORE_3 M7
VDDCORE_4 M9
VDDCORE_5 M11
VDDCORE_6 N8
VDDCORE_7 N10
VDDCORE_8 N12
VDDCORE_9 P11
VFBSMPS G2

VCORE_0V81

C15 22uF
C16 22uF
C17 22uF
C18 22uF
C69 1uF
C50 1uF
C51 1uF

GND

C52 1uF
C53 1uF
C54 1uF
C55 1uF
C56 1uF
C57 1uF
C58 1uF

VDDSMPS_1 K1
VDDSMPS_2 K2
VDDSMPS_3 K3
VDDSMPS_4 K4
VDDSMPS_5 K5
VDDSMPS_6 K6

1V8

VLXSMPS_1 J1
VLXSMPS_2 J2
VLXSMPS_3 J3
VLXSMPS_4 J4
VLXSMPS_5 J5
VLXSMPS_6 J6

MCU_SMPS_SW

L3
1uH

C59
2.2nF

MCU_SMPS_SNUBBER

R19
2ohm

V08CAP F1

V08CAP

C19
4.7uF

MCU internal SMPS

Presentation revision; electrical design review remains pending

Global X Camera

Sheet: /5 MCU internal SMPS and core supply/
File: 05_mcu_core.kicad_sch

Title: MCU internal SMPS and core supply

Size: A3	Date: 2026-09-05	**Rev: A-readable**
KiCad E.D.A. 10.0.6		Id: 5/14

Clock, reset, boot and debug

Local oscillator and strap components are wired to the MCU. Optional SW2/SW3 remain DNP.

U1D
STM32N657X0H3Q

TP3
TP3
3V3

TP4
TP4
1V8

TP5
TP5
VCORE_0V81

PWR_HOLD
E2
PWR_ON

SHUTTER_SENSE
F3
PC13

STATUS_LED_K
C10
PC0

1V8
A1
PDR_ON

PH0-OSC_IN
A7
H5E_IN

C36
8.2pF
GND

Y1
CX3225SA0000D0PTWCC
1
3
2

PH1-OSC_OUT
B7
H5E_OUT

GND

C37
8.2pF
GND

TP6
TP6

BOOT0
G4
BOOT0

SW2
B3U-3000P
1
2

1V8

R8
10kohm
GND

PA6
P8
BOOT1

R9
10kohm
GND

TP7
TP7

NRST
G3
NRST

SW3
B3U-3000P
1
2

GND

C38
100nF
GND

PA13
U6
SWDIO

PA14
T6
SWCLK

PB5
U12
SWO

J2
TC2030-IDC-NL
5
2
4
6
1
1
3
3

3V3

GND

MCU system control

SWD header

Presentation revision; electrical design review remains pending

Global X Camera

Sheet: /6 Clock, reset, boot and debug/
File: 06_control.kicad_sch

Title: Clock, reset, boot and debug

Size: A3	Date: 2026-09-05	**Rev: A-readable**
KiCad E.D.A. 10.0.6		Id: 6/14

Camera connector, module and CSI-2

One CSI-2 data lane plus clock. CAM1 represents the external module; J3 is the host-board receptacle.

U1G
STM32N657X0H3Q

CAM_MIPI: differential clock and data lane 0

{CAM_MIPI}

CSI_CKP T3 CAM_CSI_CLK_P
CSI_CKN U3 CAM_CSI_CLK_N
CSI_D0P T5 CAM_CSI_D0_P
CSI_D0N U5 CAM_CSI_D0_N

CAM_I2C: camera control

{CAM_I2C}

PC10 A8 I2C4_SCL
PC11 B8 I2C4_SDA

1V8
R22
4.7kohm
R23
4.7kohm
TP13
TP13

PC9 C7 CAM_MCLK

PC8 D7 CAM_XSHUTDOWN
R20
100kohm
GND

CSI_REXT P5 CSI_REXT
R17
200ohm
GND

PC12 C8 CAM_AVDD_EN

1V8
C66
100nF
C70
1uF
GND

MCU camera interface

J3
BM28B0.6-30DS/2-0.35V(51)

2 GND
3
8
14
20
23
26
29
30
GND_CONTACT 31
GND_CONTACT 32
GND_CONTACT 33
GND_CONTACT 34

16 CAM_CSI_CLK_P
18 CAM_CSI_CLK_N
10 CAM_CSI_D0_P
12 CAM_CSI_D0_N
25 I2C4_SCL
27 I2C4_SDA
28
6
1 CAM_VANA_2V8
4 CAM_VCORE_1V2
5

7 9 11 13 15 17 19 21 22 24

CAM1
CAM-5G0-079CLR

CAM_CSI_CLK_P 16 CLKP
CAM_CSI_CLK_N 18 CLKN
CAM_CSI_D0_P 10 DATA1P
CAM_CSI_D0_N 12 DATA1N
I2C4_SCL 25 SCL
I2C4_SDA 27 SDA
CAM_MCLK 28 CLKIN
CAM_XSHUTDOWN 6 XSHUTDOWN
CAM_VANA_2V8 1 VANA_2V8
CAM_VCORE_1V2 4 VCORE_1V15
1V8 5 VDDIO_1V8

GND 2 GND
GND 3
GND 8
GND 14
GND 20
GND 23
GND 26
GND 29
GND 30
GND 31
GND 32
GND 33
GND 34

GPIO0 7
GPIO1 9
GPIO2 11
GPIO3 13
GPIO4 15
GPIO5 17
NC 19
NC 21
DATA2P 22
DATA2N 24

Presentation revision; electrical design review remains pending

Global X Camera

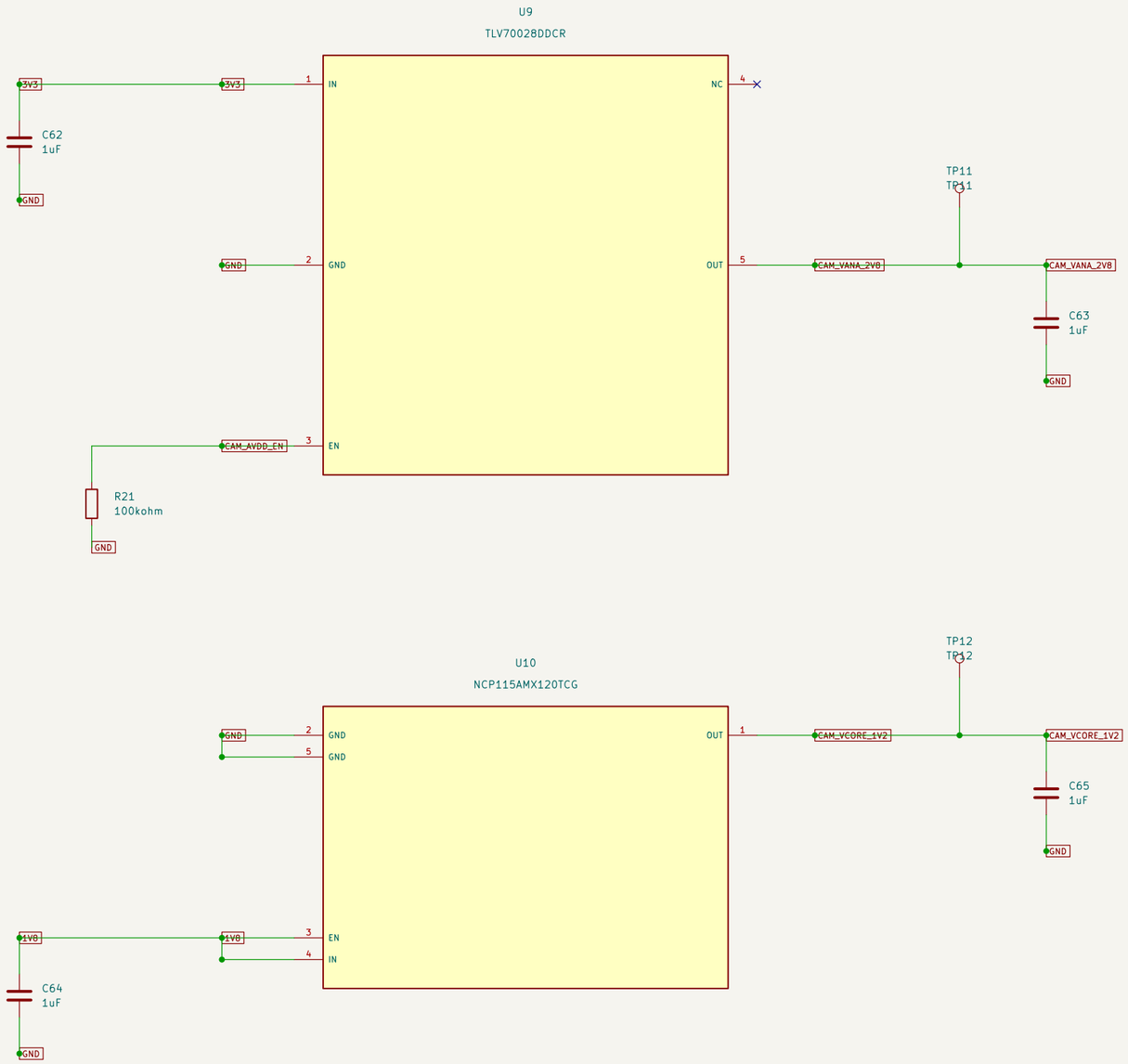
Sheet: /7 Camera connector, module and CSI-2/
File: 07_camera.kicad_sch

Title: Camera connector, module and CSI-2

Size: A3	Date: 2026-09-05	**Rev: A-readable**
KiCad E.D.A. 10.0.6		Id: 7/14

Camera power rails

Analog and core supplies, with their local input/output capacitors and enable bias.



Presentation revision; electrical design review remains pending

Global X Camera

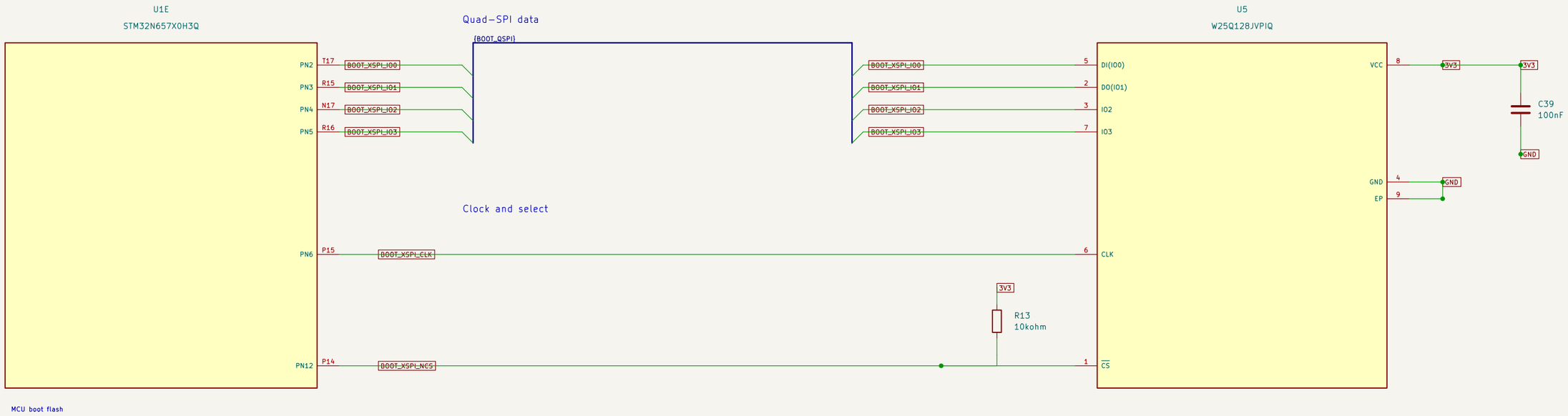
Sheet: /8 Camera power rails/
File: 08_camera_power.kicad_sch

Title: Camera power rails

Size: A3	Date: 2026-09-05	**Rev: A-readable**
KiCad E.D.A. 10.0.6		Id: 8/14

Boot NOR flash

Direct signal paths, grouped interfaces and adjacent support components.



Presentation revision; electrical design review remains pending

Global X Camera

Sheet: /9 Boot NOR flash/
File: 09_boot_flash.kicad_sch

Title: Boot NOR flash

Size: A3	Date: 2026-09-05	**Rev: A-readable**
KiCad E.D.A. 10.0.6		Id: 9/14

Raw NAND storage

Direct signal paths, grouped interfaces and adjacent support components.

U1F

STM32N657X0H3Q

8-bit data

{NAND_DATA}

U6

MX30LF2G18AC-TI

PA12 U7 NAND_D0

PA11 T7 NAND_D1

PA10 R7 NAND_D2

PA9 P7 NAND_D3

PA8 U8 NAND_D4

PA2 T8 NAND_D5

PA1 P9 NAND_D6

PA0 U10 NAND_D7

NAND_D0 29 IO0

NAND_D1 30 IO1

NAND_D2 31 IO2

NAND_D3 32 IO3

NAND_D4 41 IO4

NAND_D5 42 IO5

NAND_D6 43 IO6

NAND_D7 44 IO7

FMC control

PG10 T10 NAND_CLE 16 CLE

PA3 R14 NAND_ALE 17 ALE

PF8 N3 NAND_NWE 18 WE

PA5 R8 NAND_NOE 8 RE

3V3

R15

10kohm

PG14 P6 NAND_NCE 9 CE

3V3

R14

10kohm

PF2 N4 NAND_RNB 7 R/B

GND

R16

10kohm

PA4 U14 NAND_NWP 19 WP

VCC 12 3V3

VCC1 34

VCC 37

VCC1 39

3V3

C40

100nF

C41

100nF

GND

3V3

C42

100nF

C71

100nF

GND

VSS 13 GND

VSS1 25

VSS 36

VSS1 48

PT 38 NAND_PT

R27

100kohm

GND

NC 1

NC 2

NC 3

NC 4

NC 5

NC 6

NC 10

NC 11

NC 14

NC 15

NC 20

NC 21

NC 22

NC 23

NC 24

NC 26

NC 27

NC 28

NC 33

NC 35

NC 40

NC 45

NC 46

NC 47

MCU nand

Presentation revision; electrical design review remains pending

Global X Camera

Sheet: /10 Raw NAND storage/

File: 10_nand.kicad_sch

Title: Raw NAND storage

Size: A3	Date: 2026-09-05	**Rev: A-readable**
KiCad E.D.A. 10.0.6		Id: 10/14

USB-C data, protection and supply

USB data pair is routed as one functional group; connector CC resistors and protection remain local.

U1H
STM32N657X0H3Q
OTG1_HS_DP
B5
USB_DP
OTG1_HS_DM
A5
USB_DM
OTG1_TX_RTUNE
C6
USB_TX_RTUNE
MCU USB HS
R18
200ohm
GND
D1
USBLC6-2SC6
USB_DP
1
I/O1
6
I/O1
USB_DM
3
I/O2
4
I/O2
VBUS
5
USB_VBUS
GND
2
GND
USB ESD array
C43
4.7uF
GND
J1
USB4105-GF-A
A6
DP1
B6
DP2
A7
DN1
B7
DN2
USB_CC1
A5
CC1
R10
5.1kohm
GND
USB_CC2
B5
CC2
R11
5.1kohm
GND
USB_VBUS
A4_B9
VBUS
B4_A9
VBUS_1
SBU1
A8
SBU2
B8
GND
A1_B12
GND
GND_1
B1_A12
SHELL_GND
SH1
SHELL_GND_1
SH2
SHELL_GND_2
SH3
SHELL_GND_3
SH4
USB-C receptacle
D2
SS34
SYS_IN
U4
TLV70033DDCT
USB_VBUS
1
IN
3
EN
GND
2
GND
OUT
5
3V3_USB
N/C
4
3V3_USB
C13
1uF
GND
C14
1uF
GND
USB analog supply

Presentation revision; electrical design review remains pending

Global X Camera

Sheet: /11 USB-C data, protection and supply/
File: 11_usb.kicad_sch

Title: USB-C data, protection and supply

Size: A3	Date: 2026-09-05	**Rev: A-readable**
KiCad E.D.A. 10.0.6		Id: 11/14

RTC and accelerometer interfaces

Separate MCU buses for the RTC and switched accelerometer. Bias and bypass networks are local to their ICs.

U1I
STM32N657X0H3Q
PE5
C11
RTC_I2C1_SCL
PE6
D11
RTC_I2C1_SDA
{RTC_I2C}
PH2
E7
RTC_INT_N
PC2
D9
ACCEL_I2C_SCL
PC3
E10
ACCEL_I2C_SDA
{ACCEL_I2C}
PC4
A10
ACCEL_INT1
PC1
A9
ACCEL_PWR_EN
MCU RTC / orientation
3V3
R28
4.7kohm
R29
4.7kohm
R26
1Mohm
GND
U8
RV-3028-C8-32.768kHz-1PPM-TA-QC
SCL
SDA
INT
CLKOUT
VSS
VDD
VBACKUP
EVI
C67
100nF
TP10
TP10
BAT_CELL1_TAP
C68
100nF
R25
10kohm
RTC_EVI
R24
10kohm
TP8
TP8
TP9
TP9
ACCEL_1V8_SW
R31
10kohm
R30
10kohm
U11
LIS2DW12TR
SCL/SPC
SDA/SDI/SDO
VDD
VDDIO
CS
SDO/SA0
GND
RES
GND
INT1
NC
INT2
C72
100nF
C73
1uF
U12
TPS22918DBVT
ON
VIN
GND
VOUT
QOD
CT
1V8
R32
100kohm
C74
1uF
Accelerometer load switch

Presentation revision; electrical design review remains pending

Global X Camera

Sheet: /12 RTC and accelerometer interfaces/
File: 12_rtc_accelerometer.kicad_sch

Title: RTC and accelerometer interfaces

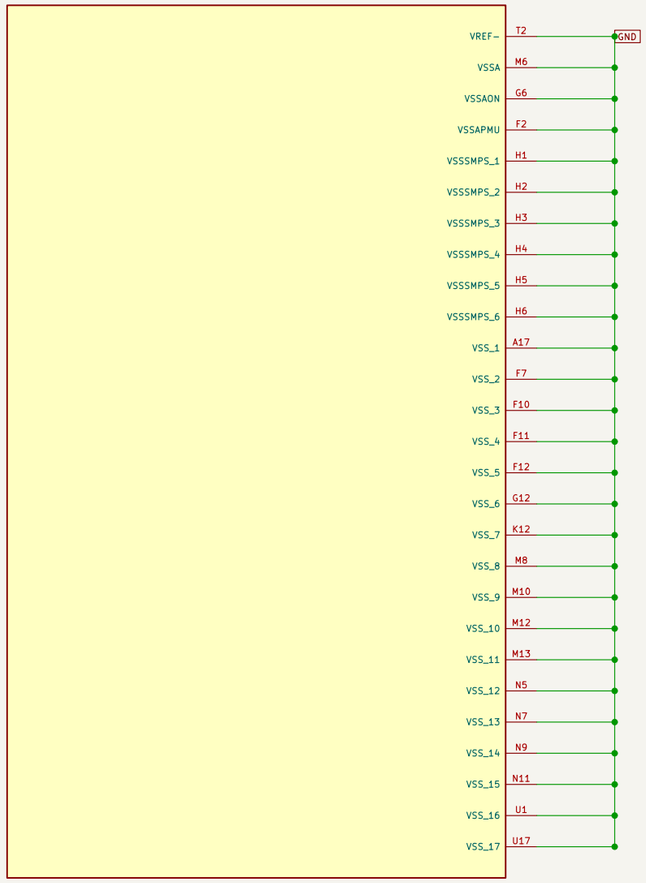
Size: A3	Date: 2026-09-05	**Rev: A-readable**
KiCad E.D.A. 10.0.6		Id: 12/14

MCU grounds and unused GPIO A–D

All physical ground balls and unused pins remain individually numbered. NC marks preserve source no-connect intent.

U1C

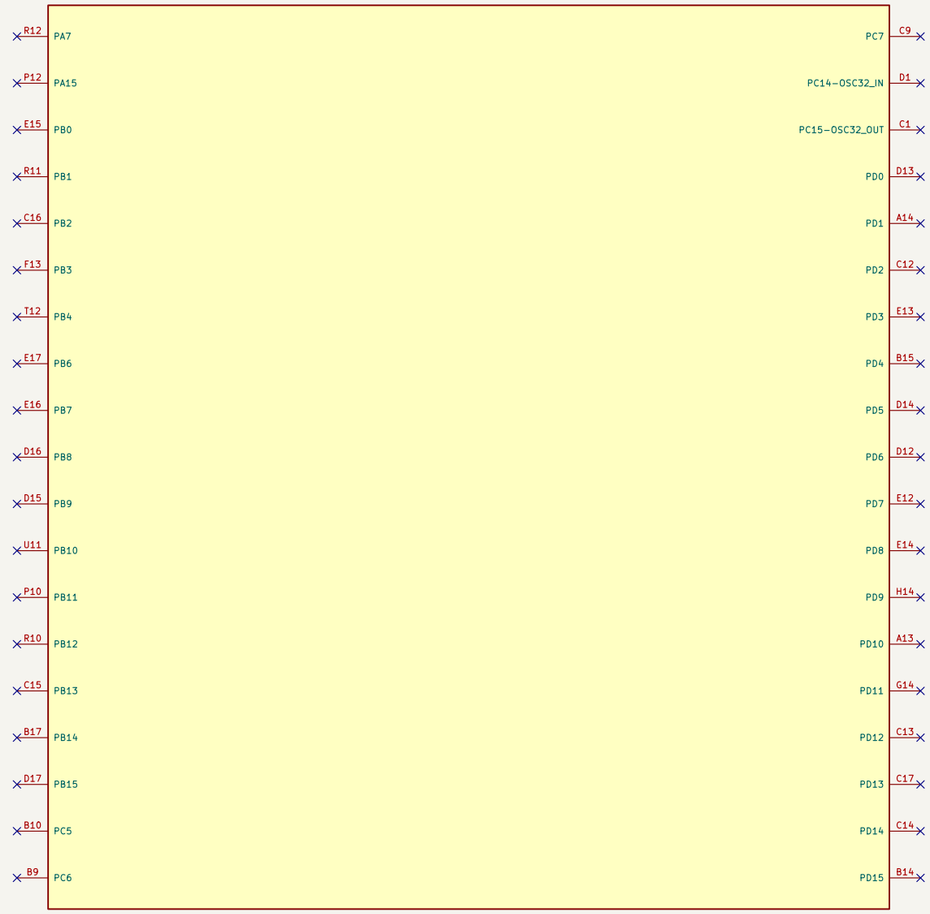
STM32N657X0H3Q



MCU ground balls

U1J

STM32N657X0H3Q



Unused GPIO A–D

Presentation revision; electrical design review remains pending

Global X Camera

Sheet: /13 MCU grounds and unused GPIO A–D/

File: 13_ground_unused.kicad_sch

Title: MCU grounds and unused GPIO A–D

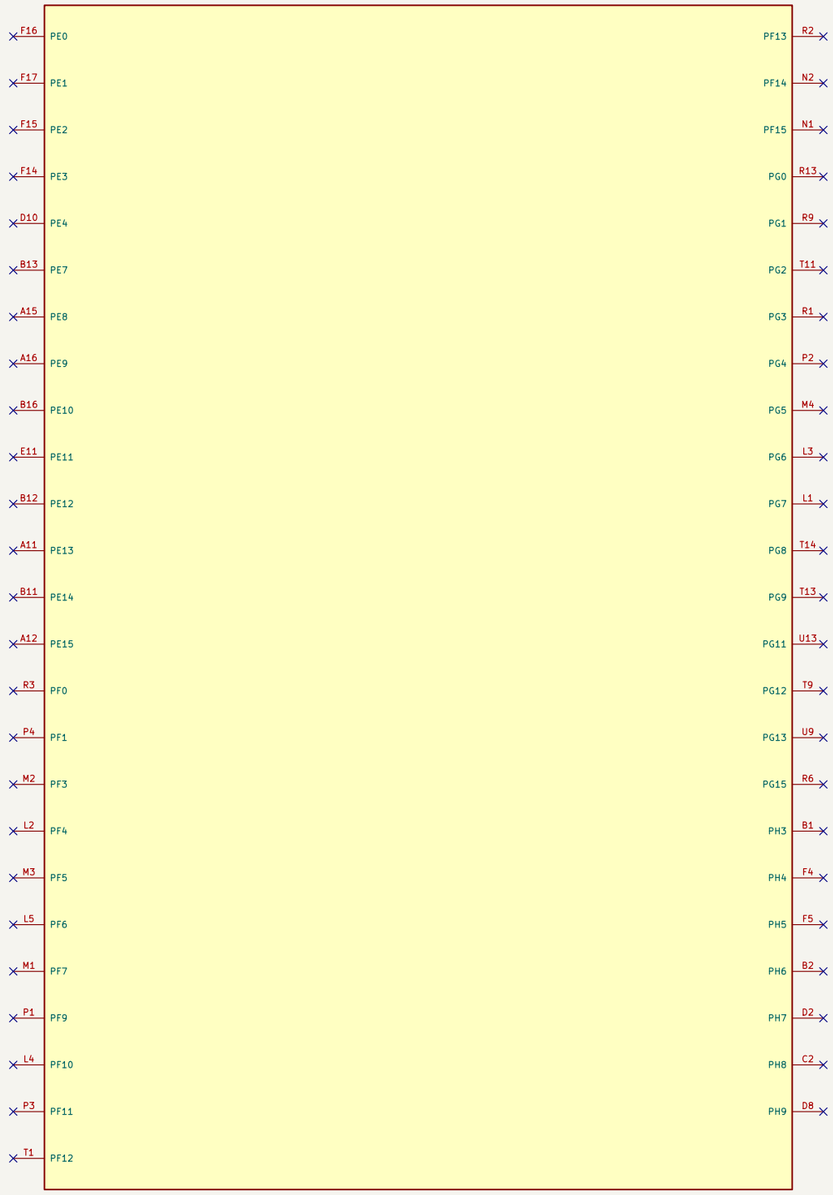
Size: A3	Date: 2026-09-05	**Rev: A-readable**
KiCad E.D.A. 10.0.6		Id: 13/14

Unused MCU GPIO and interfaces

Explicit no-connects retained for every unused MCU terminal. These pages are reference appendices.

U1K

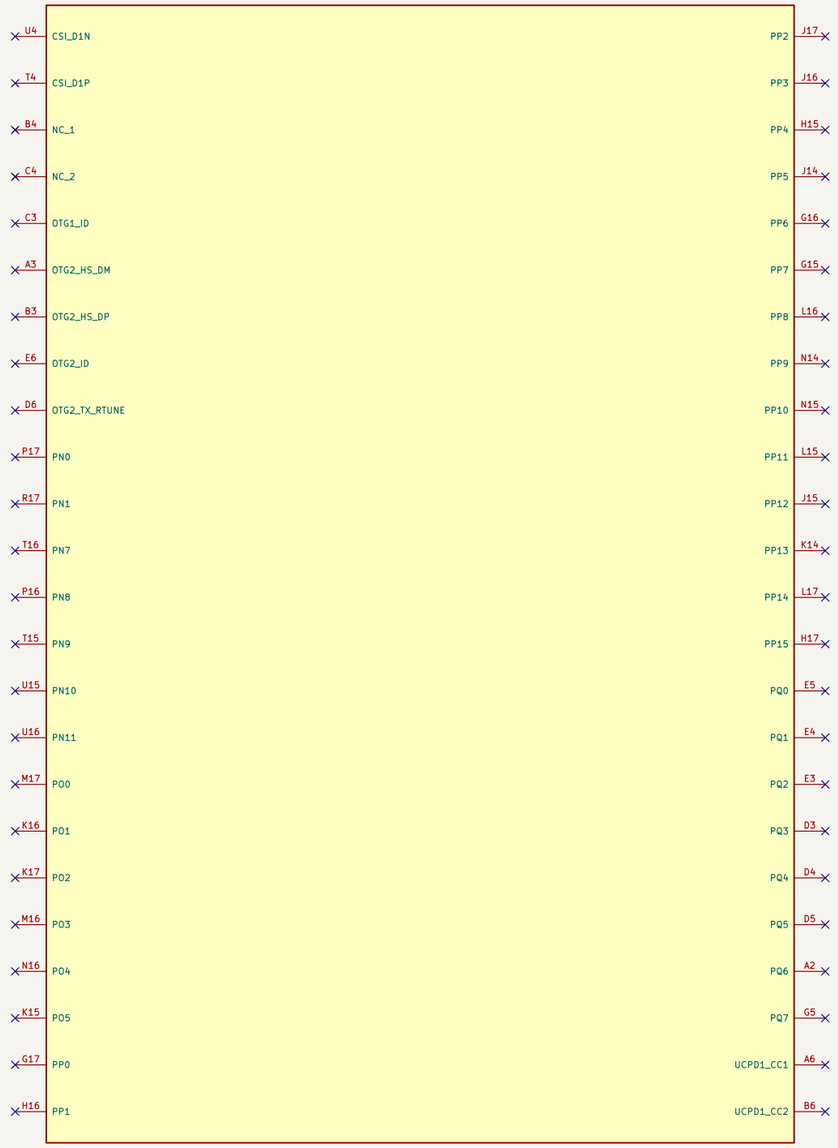
STM32N657X0H3Q



Unused MCU terminals

U1L

STM32N657X0H3Q



Unused MCU terminals

Presentation revision; electrical design review remains pending

Global X Camera

Sheet: /14 Unused MCU GPIO and interfaces/
File: 14_unused_gpio.kicad_sch

Title: Unused MCU GPIO and interfaces

Size: A3	Date: 2026-09-05	**Rev: A-readable**
KiCad E.D.A. 10.0.6		Id: 14/14